Supplementary Material for Which Classicality? Incidence, Simplex, and Product-Rule Tests in Finite Quantum Logics

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1 Scope and archive structure

This supplement specifies the source coordinates, validation rules, factorization problem, numerical results, and certificate status for every row reported in the manuscript. One program, `simplex_audit_rebuilt.py`, performs the complete calculation. Its default action validates all sources, runs every projector-cone task, runs both operational closures, and evaluates the completed KCBS pentagon. Per-task SHA-256 input fingerprints permit exact checkpoint reuse; `--force` requests a fresh evaluation.

The executed program has SHA-256 digest

`d2298aeaab2d47a42e94e9077ece208df8042ddf6603d6fd0252d8fc184e1596.`

The digest is recorded in `audit_v3/audit_report.json`. The run used CPython 3.11.9 on 64-bit Windows, completed all requested tasks, and recorded no run errors.

The archive contains:

`simplex_audit_rebuilt.py` Source-driven validation and audit program.

`audit_v3/audit_report.json` Consolidated runtime, validation, projector, closure, pentagon, and cache report.

`audit_v3/vectors.json` Machine-readable representatives, labels, contexts, sources, and valuation checks.

`audit_v3/projector/` Per-configuration projector reports and all available exact certificates.

`audit_v3/closures/` Both closure reports for every rational or quadratic source configuration.

`audit_v3/pentagon/` Numerical completed-pentagon report.

`2026-simplex-vector-catalog.tex` Complete explicit ray and context catalogue printed in Sec. 7.

`2026-simplex-supplement.tex/pdf` This document in source and compiled form.

`README.md` Commands, dependencies, file map, and status rules.

`SHA256SUMS.txt` Digest and relative path for every payload file.

Table 1: Validated configurations. The valuation column gives the number of two-valued states when that enumeration is part of the source-property check.

configuration	rays	contexts	valuations	validation property
Firefly L_{12}	5	2	–	projective geometry
Yu–Oh 13	13	4	–	published seed
Yu–Oh 25	25	16	–	orthogonal completion
Specker bug	13	7	14	distinguished nonfull pair
Kochen–Specker Γ_3	27	17	24	distinguished equal pair
Tkadlec nonunital	37	26	8	eight never-true rays
Cabello 18–9	18	9	0	Kochen–Specker set
Peres–Mermin 24	24	24	0	operator-origin check

2 Primary coordinate resources

The Specker bug and the Kochen–Specker Γ_3 configuration are sourced from the coordinate-labelled diagrams in Tkadlec's treatment and the Kochen–Specker construction itself. Cabello gives an independent parametric coordinatization of the Bell/Specker TIFS family. The remaining sources and construction rules are listed in Table 1. Full bibliographic data appear at the end of this supplement.

For the Specker bug, the coordinates lie in $\mathbb{Q}(\sqrt{2})$ but the complete Born table is rational. The audit therefore constructs an exact rational rank factorization before facet enumeration. For Γ_3, the coordinate-labelled displayed rays are orthogonally completed to 27 rays and 17 triads, as required by the full suborthoposet. Some Born overlaps remain in $\mathbb{Q}(\sqrt{2}) \setminus \mathbb{Q}$, so its facet and LP branch is numerical.

The Tkadlec nonunital construction begins with the 25 marked rays and adjoins every distinct cross-product ray completing an orthogonal pair. Exact integer geometry gives 37 rays and 26 triads. Direct enumeration yields eight two-valued states and eight never-true rays.

3 Implemented orthant-factorization problem

Real symmetric operators are represented by coordinates whose trace metric is

$$G = \operatorname{diag}(\underbrace{1, \dots, 1}_{d}, \underbrace{2, \dots, 2}_{d(d-1)/2}).$$

Effects are stored in dual coordinates $\widetilde{I}_E = GI_E$. After quotienting operational null directions and reducing to accessible coordinates, the program computes homogeneous facet matrices H_Ω, H_E and solves

$$H_E^T \sigma H_\Omega = \widetilde{I}_E^T \big[(1-r)\,\mathrm{Id} + rD\big] I_\Omega, \qquad \sigma \geq 0, \qquad 0 \leq r \leq 1,$$

where D maps every normalized preparation to the maximally mixed state. For a general operational rank factorization of a Born table, the same equation is written in the induced rational or numerical coordinates.

The associated ontological normalization requirements are

$$\sum_\lambda \mu_\omega(\lambda) = 1, \qquad \xi_u(\lambda) = 1, \qquad 0 \leq \xi_e(\lambda) \leq 1.$$

Thus the implemented test is the general finite orthant-factorization test, not an assumption that an intermediate simplicial cone has the accessible-space dimension.

3.1 Exact certificate convention

Put

$$M_0 = \widetilde{I}_E^T I_\Omega, \qquad M_1 = \widetilde{I}_E^T (D - \mathrm{Id}) I_\Omega,$$

and let $C_{\ell k} = (h_\ell^E)^T h_k^\Omega$. An exact primal certificate is a rational nonnegative matrix σ satisfying

$$\sum_{\ell,k} \sigma_{\ell k} C_{\ell k} - r M_1 = M_0.$$

An exact dual certificate is a rational matrix Y satisfying

$$\langle Y, C_{\ell k} \rangle \le 0, \qquad -\langle Y, M_1 \rangle \le 1, \qquad \langle Y, M_0 \rangle = r.$$

The label "exact optimum" is used only when both certificates pass exact rational verification. A decimal approximation or a rationalization of a floating-point optimum is labelled numerical.

4 Projector-cone results

Table 2: Projector-cone audit. Facets are listed as state/effect counts.

configuration	facets	r	status
Firefly L_{12}	5/5	0	exact optimum
Completed pentagon	12/12	0.119140568134	numerical
Specker bug	26/26	25/79	exact optimum
Kochen–Specker Γ_3	230/230	0.502197722972	numerical
Tkadlec nonunital	356/356	0.547491649818	numerical
Yu–Oh 13	24/24	3/8	exact optimum
Yu–Oh 25	96/96	21/44	exact optimum
Cabello 18–9	146/146	1/3	exact optimum
Peres–Mermin 24	120/120	4/9	exact optimum

The Specker-bug certificate verifies $r = 25/79$ exactly. The Γ_3 algebraic branch has maximum dual violation 1.6×10^{-10}. The Tkadlec nonunital equality residual is 2.8×10^{-13}. The completed-pentagon residual is 1.8×10^{-13}. No exact value is assigned to these three numerical rows.

5 Vector-generated operational closures

The closed-effects variant starts from the ray preparations and the maximally mixed state and closes the effects under the unit, residual sharp outcomes, complements, coarse grainings, and operator equality. The effects-and-preps variant also adds every normalized closed effect as a preparation. This final step is an explicit modelling choice.

Table 3: Closure audit. Generator and facet columns list state/effect pairs. Every threshold in this table is numerical.

configuration	variant	generators	facets	r_{num}	solver	status
Firefly L_{12}	effects	6/11	5/5	0	primal	numerical
Firefly L_{12}	effects+preps	11/11	5/5	0	primal	numerical
Yu–Oh 13	effects	14/51	24/96	0.450000000000	primal	numerical
Yu–Oh 13	effects+preps	51/51	96/96	0.477272727273	primal	numerical
Yu–Oh 25	effects	26/51	96/96	0.477272727273	primal	numerical
Yu–Oh 25	effects+preps	51/51	96/96	0.477272727273	primal	numerical
Specker bug	effects	14/27	26/26	0.316455696203	primal	numerical
Specker bug	effects+preps	27/27	26/26	0.316455696203	primal	numerical
Kochen–Specker Γ_3	effects	28/55	230/230	0.502197722951	cutting dual	numerical
Kochen–Specker Γ_3	effects+preps	55/55	230/230	0.502197722951	cutting dual	numerical
Tkadlec nonunital	effects	38/75	356/356	0.547491649818	cutting dual	numerical
Tkadlec nonunital	effects+preps	75/75	356/356	0.547491649818	cutting dual	numerical
Cabello 18–9	effects	19/121	146/120	0.444444444444	primal	numerical
Cabello 18–9	effects+preps	121/121	120/120	0.444444444444	primal	numerical
Peres–Mermin 24	effects	25/139	120/120	0.444444444444	primal	numerical
Peres–Mermin 24	effects+preps	139/139	120/120	0.444444444444	primal	numerical

The Specker-bug closures agree numerically with 25/79. The Γ_3 closures agree with each other to the displayed precision and have maximum cutting-dual violations below 7×10^{-15}. The two Tkadlec nonunital violations are 7.1×10^{-11} and 5.9×10^{-12}. Fractions suggested by numerical closure values are comparisons only and are not exact closure certificates.

6 Reproduction and checkpoint use

With NumPy, SciPy, SymPy, and a rational/GMP-enabled pycddlib installation, the complete command is

```
python simplex_audit_rebuilt.py --outdir audit_v3
```

The same command may be issued after an interruption or on a populated output directory. A task is reused only when its stored SHA-256 input fingerprint and audit-schema version match. The run log prints `REUSE` for such a task. To request every calculation explicitly, use

```
python simplex_audit_rebuilt.py --outdir audit_v3 --force
```

Validation-only and selected-configuration runs are available through `--validate` and repeatable `--only` arguments.

7 Complete vector and context catalogue

Each ray is projective: any nonzero scalar multiple denotes the same ray. The ordering below is exactly the ordering stored in `audit_v3/vectors.json`. Context indices are one-based in this catalogue.

7.1 Firefly logic L_{12}

Source: D. W. Cohen, An Introduction to Hilbert Space and Quantum Logic, Springer (1989), DOI 10.1007/978-1-4613-8841-8. Construction: explicit five-ray firefly realization. The audit verifies 5 projectively distinct rays and 2 complete contexts.

ray	representative
v_1	$(1, 0, 0)$
v_2	$(0, 1, 0)$
v_3	$(0, 0, 1)$
v_4	$(0, 1, 1)$
v_5	$(0, 1, -1)$

context	mutually orthogonal rays
C_1	$\{v_1, v_2, v_3\}$
C_2	$\{v_1, v_4, v_5\}$

7.2 Yu–Oh 13

Source: S. Yu and C. H. Oh, Phys. Rev. Lett. 108, 030402 (2012), DOI 10.1103/PhysRevLett.108.030402. Construction: published 13-ray seed. The audit verifies 13 projectively distinct rays and 4 complete contexts.

ray	representative
v_1	$(1, 0, 0)$
v_2	$(0, 1, 0)$
v_3	$(0, 0, 1)$
v_4	$(0, 1, 1)$
v_5	$(0, 1, -1)$
v_6	$(1, 0, 1)$
v_7	$(1, 0, -1)$
v_8	$(1, 1, 0)$
v_9	$(1, -1, 0)$
v_{10}	$(1, 1, 1)$
v_{11}	$(1, 1, -1)$
v_{12}	$(1, -1, 1)$
v_{13}	$(1, -1, -1)$

context	mutually orthogonal rays
C_1	$\{v_1, v_2, v_3\}$
C_2	$\{v_1, v_4, v_5\}$
C_3	$\{v_2, v_6, v_7\}$
C_4	$\{v_3, v_8, v_9\}$

7.3 Yu–Oh 25-ray orthogonal completion

Source: S. Yu and C. H. Oh, Phys. Rev. Lett. 108, 030402 (2012), DOI 10.1103/PhysRevLett.108.030402. Construction: cross-product completion of every orthogonal pair in the 13-ray seed. The audit verifies 25 projectively distinct rays and 16 complete contexts.

ray	representative
v_1	$(0, 0, 1)$
v_2	$(0, 1, -1)$
v_3	$(0, 1, 0)$
v_4	$(0, 1, 1)$
v_5	$(1, -1, -1)$
v_6	$(1, -1, -2)$
v_7	$(1, -1, 0)$
v_8	$(1, -1, 1)$
v_9	$(1, -1, 2)$
v_{10}	$(1, -2, -1)$
v_{11}	$(1, -2, 1)$
v_{12}	$(1, 0, -1)$
v_{13}	$(1, 0, 0)$
v_{14}	$(1, 0, 1)$
v_{15}	$(1, 1, -1)$
v_{16}	$(1, 1, -2)$
v_{17}	$(1, 1, 0)$
v_{18}	$(1, 1, 1)$
v_{19}	$(1, 1, 2)$
v_{20}	$(1, 2, -1)$
v_{21}	$(1, 2, 1)$
v_{22}	$(2, -1, -1)$

ray	representative
v_{23}	$(2,-1,1)$
v_{24}	$(2,1,-1)$
v_{25}	$(2,1,1)$

context	mutually orthogonal rays
C_1	$\{v_1, v_3, v_{13}\}$
C_2	$\{v_1, v_7, v_{17}\}$
C_3	$\{v_2, v_4, v_{13}\}$
C_4	$\{v_2, v_5, v_{25}\}$
C_5	$\{v_2, v_{18}, v_{22}\}$
C_6	$\{v_3, v_{12}, v_{14}\}$
C_7	$\{v_4, v_8, v_{24}\}$
C_8	$\{v_4, v_{15}, v_{23}\}$
C_9	$\{v_5, v_9, v_{17}\}$
C_{10}	$\{v_5, v_{14}, v_{20}\}$
C_{11}	$\{v_6, v_8, v_{17}\}$
C_{12}	$\{v_7, v_{15}, v_{19}\}$
C_{13}	$\{v_7, v_{16}, v_{18}\}$
C_{14}	$\{v_8, v_{12}, v_{21}\}$
C_{15}	$\{v_{10}, v_{14}, v_{15}\}$
C_{16}	$\{v_{11}, v_{12}, v_{18}\}$

7.4 Specker bug

Source: J. Tkadlec, Greechie Diagrams of Small Quantum Logics with Small State Spaces, Int. J. Theor. Phys. 37, 203–209 (1998), Fig. 1a, DOI 10.1023/A:1026646229896; A. Cabello Quintero, Pruebas algebraicas de imposibilidad de variables ocultas en mecanica cuantica, PhD thesis, Universidad Complutense de Madrid (1996), Fig. 2.5 and printed p. 44. Construction: coordinate-labelled 13-line realization; Cabello supplies the parametric B-11/B-13 family. The audit verifies 13 projectively distinct rays and 7 complete contexts.

ray	representative
v_1	$(1,\sqrt{2},1)$
v_2	$(1,0,-1)$
v_3	$(1,-\sqrt{2},1)$
v_4	$(1,\sqrt{2},-1)$
v_5	$(1,0,1)$
v_6	$(1,-\sqrt{2},-1)$
v_7	$(1,-1/2\,\sqrt{2},0)$
v_8	$(1,1/2\,\sqrt{2},0)$
v_9	$(0,1,0)$
v_{10}	$(1,\sqrt{2},-3)$
v_{11}	$(1,\sqrt{2},3)$
v_{12}	$(1,-\sqrt{2},-3)$
v_{13}	$(1,-\sqrt{2},3)$

context	mutually orthogonal rays
C_1	$\{v_1, v_2, v_3\}$
C_2	$\{v_1, v_7, v_{10}\}$
C_3	$\{v_2, v_5, v_9\}$
C_4	$\{v_3, v_8, v_{12}\}$
C_5	$\{v_4, v_5, v_6\}$
C_6	$\{v_4, v_7, v_{11}\}$
C_7	$\{v_6, v_8, v_{13}\}$

7.5 Kochen–Specker Γ_3

Source: J. Tkadlec, Greechie Diagrams of Small Quantum Logics with Small State Spaces, Int. J. Theor. Phys. 37, 203–209 (1998), Fig. 1b, DOI 10.1023/A:1026646229896. Construction: orthogonal completion of the coordinate-labelled displayed rays. The audit verifies 27 projectively distinct rays and 17 complete contexts.

ray	representative
v_1	$(0, 0, 1)$
v_2	$(0, 1, -1)$
v_3	$(0, 1, 0)$
v_4	$(0, 1, 1)$
v_5	$(1, -1/2\sqrt{2}, -1/2\sqrt{2})$
v_6	$(1, -1/2\sqrt{2}, -3/2\sqrt{2})$
v_7	$(1, -1/2\sqrt{2}, 0)$
v_8	$(1, -1/2\sqrt{2}, 1/2\sqrt{2})$
v_9	$(1, -1/2\sqrt{2}, 3/2\sqrt{2})$
v_{10}	$(1, -\sqrt{2}, -1)$
v_{11}	$(1, -\sqrt{2}, -3)$
v_{12}	$(1, -\sqrt{2}, 0)$
v_{13}	$(1, -\sqrt{2}, 1)$
v_{14}	$(1, -\sqrt{2}, 3)$
v_{15}	$(1, 0, -1)$
v_{16}	$(1, 0, 0)$
v_{17}	$(1, 0, 1)$
v_{18}	$(1, 1/2\sqrt{2}, -1/2\sqrt{2})$
v_{19}	$(1, 1/2\sqrt{2}, -3/2\sqrt{2})$
v_{20}	$(1, 1/2\sqrt{2}, 0)$
v_{21}	$(1, 1/2\sqrt{2}, 1/2\sqrt{2})$
v_{22}	$(1, 1/2\sqrt{2}, 3/2\sqrt{2})$
v_{23}	$(1, \sqrt{2}, -1)$
v_{24}	$(1, \sqrt{2}, -3)$
v_{25}	$(1, \sqrt{2}, 0)$
v_{26}	$(1, \sqrt{2}, 1)$
v_{27}	$(1, \sqrt{2}, 3)$

context	mutually orthogonal rays
C_1	$\{v_1, v_3, v_{16}\}$
C_2	$\{v_1, v_7, v_{25}\}$
C_3	$\{v_1, v_{12}, v_{20}\}$

context	mutually orthogonal rays
C_4	$\{v_2, v_4, v_{16}\}$
C_5	$\{v_2, v_5, v_{21}\}$
C_6	$\{v_3, v_{15}, v_{17}\}$
C_7	$\{v_4, v_8, v_{18}\}$
C_8	$\{v_5, v_9, v_{25}\}$
C_9	$\{v_6, v_8, v_{25}\}$
C_{10}	$\{v_7, v_{23}, v_{27}\}$
C_{11}	$\{v_7, v_{24}, v_{26}\}$
C_{12}	$\{v_{10}, v_{14}, v_{20}\}$
C_{13}	$\{v_{10}, v_{17}, v_{23}\}$
C_{14}	$\{v_{11}, v_{13}, v_{20}\}$
C_{15}	$\{v_{12}, v_{18}, v_{22}\}$
C_{16}	$\{v_{12}, v_{19}, v_{21}\}$
C_{17}	$\{v_{13}, v_{15}, v_{26}\}$

7.6 Tkadlec nonunital configuration

Source: J. Tkadlec, Greechie Diagrams of Small Quantum Logics with Small State Spaces, Int. J. Theor. Phys. 37, 203–209 (1998), Fig. 2, DOI 10.1023/A:1026646229896. Construction: orthogonal completion of the 25 rays marked in Fig. 2. The audit verifies 37 projectively distinct rays and 26 complete contexts.

ray	representative
v_1	$(0, 0, 1)$
v_2	$(0, 1, -1)$
v_3	$(0, 1, 0)$
v_4	$(0, 1, 1)$
v_5	$(1, -1, -1)$
v_6	$(1, -1, -2)$
v_7	$(1, -1, 0)$
v_8	$(1, -1, 1)$
v_9	$(1, -1, 2)$
v_{10}	$(1, -2, -1)$
v_{11}	$(1, -2, 1)$
v_{12}	$(1, -5, -2)$
v_{13}	$(1, -5, 2)$
v_{14}	$(1, 0, -1)$
v_{15}	$(1, 0, -2)$
v_{16}	$(1, 0, 0)$
v_{17}	$(1, 0, 1)$
v_{18}	$(1, 0, 2)$
v_{19}	$(1, 1, -1)$
v_{20}	$(1, 1, -2)$
v_{21}	$(1, 1, 0)$
v_{22}	$(1, 1, 1)$
v_{23}	$(1, 1, 2)$
v_{24}	$(1, 2, -1)$
v_{25}	$(1, 2, 1)$
v_{26}	$(1, 5, -2)$
v_{27}	$(1, 5, 2)$

ray	representative
v_{28}	$(2,-1,-1)$
v_{29}	$(2,-1,1)$
v_{30}	$(2,-5,-1)$
v_{31}	$(2,-5,1)$
v_{32}	$(2,0,-1)$
v_{33}	$(2,0,1)$
v_{34}	$(2,1,-1)$
v_{35}	$(2,1,1)$
v_{36}	$(2,5,-1)$
v_{37}	$(2,5,1)$

context	mutually orthogonal rays
C_1	$\{v_1, v_3, v_{16}\}$
C_2	$\{v_1, v_7, v_{21}\}$
C_3	$\{v_2, v_4, v_{16}\}$
C_4	$\{v_2, v_5, v_{35}\}$
C_5	$\{v_2, v_{22}, v_{28}\}$
C_6	$\{v_3, v_{14}, v_{17}\}$
C_7	$\{v_3, v_{15}, v_{33}\}$
C_8	$\{v_3, v_{18}, v_{32}\}$
C_9	$\{v_4, v_8, v_{34}\}$
C_{10}	$\{v_4, v_{19}, v_{29}\}$
C_{11}	$\{v_5, v_9, v_{21}\}$
C_{12}	$\{v_5, v_{17}, v_{24}\}$
C_{13}	$\{v_6, v_8, v_{21}\}$
C_{14}	$\{v_6, v_{26}, v_{33}\}$
C_{15}	$\{v_7, v_{19}, v_{23}\}$
C_{16}	$\{v_7, v_{20}, v_{22}\}$
C_{17}	$\{v_8, v_{14}, v_{25}\}$
C_{18}	$\{v_9, v_{27}, v_{32}\}$
C_{19}	$\{v_{10}, v_{17}, v_{19}\}$
C_{20}	$\{v_{11}, v_{14}, v_{22}\}$
C_{21}	$\{v_{12}, v_{20}, v_{33}\}$
C_{22}	$\{v_{13}, v_{23}, v_{32}\}$
C_{23}	$\{v_{15}, v_{29}, v_{37}\}$
C_{24}	$\{v_{15}, v_{31}, v_{35}\}$
C_{25}	$\{v_{18}, v_{28}, v_{36}\}$
C_{26}	$\{v_{18}, v_{30}, v_{34}\}$

7.7 Cabello 18–9

Source: A. Cabello, J. M. Estebaranz, and G. Garcia-Alcaine, Phys. Lett. A 212, 183–187 (1996), DOI 10.1016/0375-9601(96)00134-X. Construction: published real 18-ray coordinatization. The audit verifies 18 projectively distinct rays and 9 complete contexts.

ray	representative
v_1	$(1,0,0,0)$
v_2	$(0,0,1,-1)$
v_3	$(0,0,1,1)$

ray	representative
v_4	$(0, 1, 0, 0)$
v_5	$(0, 1, -1, 0)$
v_6	$(0, 0, 0, 1)$
v_7	$(0, 1, 1, 0)$
v_8	$(1, -1, -1, -1)$
v_9	$(1, 1, 1, -1)$
v_{10}	$(1, 0, 0, 1)$
v_{11}	$(0, 1, 0, -1)$
v_{12}	$(1, 1, -1, 1)$
v_{13}	$(1, 0, 1, 0)$
v_{14}	$(1, 1, 1, 1)$
v_{15}	$(1, 0, -1, 0)$
v_{16}	$(1, -1, 1, -1)$
v_{17}	$(1, -1, 0, 0)$
v_{18}	$(1, 1, -1, -1)$

context	mutually orthogonal rays
C_1	$\{v_1, v_2, v_3, v_4\}$
C_2	$\{v_1, v_5, v_6, v_7\}$
C_3	$\{v_2, v_{14}, v_{17}, v_{18}\}$
C_4	$\{v_3, v_9, v_{12}, v_{17}\}$
C_5	$\{v_4, v_6, v_{13}, v_{15}\}$
C_6	$\{v_5, v_8, v_9, v_{10}\}$
C_7	$\{v_7, v_{10}, v_{16}, v_{18}\}$
C_8	$\{v_8, v_{11}, v_{12}, v_{13}\}$
C_9	$\{v_{11}, v_{14}, v_{15}, v_{16}\}$

7.8 Peres–Mermin/Peres 24

Source: D. N. Mermin, Phys. Rev. Lett. 65, 3373–3377 (1990), DOI 10.1103/PhysRevLett.65.3373; A. Peres, J. Phys. A 24, L175–L178 (1991), DOI 10.1088/0305-4470/24/4/003. Construction: exact simultaneous diagonalization of the six Peres–Mermin contexts. The audit verifies 24 projectively distinct rays and 24 complete contexts.

ray	representative
v_1	$(0, 0, 0, 1)$
v_2	$(0, 0, 1, 0)$
v_3	$(0, 1, 0, 0)$
v_4	$(1, 0, 0, 0)$
v_5	$(1, -1, -1, 1)$
v_6	$(1, -1, 1, -1)$
v_7	$(1, 1, -1, -1)$
v_8	$(1, 1, 1, 1)$
v_9	$(1, -1, -1, -1)$
v_{10}	$(1, -1, 1, 1)$
v_{11}	$(1, 1, -1, 1)$
v_{12}	$(1, 1, 1, -1)$
v_{13}	$(0, 0, 1, -1)$
v_{14}	$(0, 0, 1, 1)$

ray	representative
v_{15}	$(1,-1,0,0)$
v_{16}	$(1,1,0,0)$
v_{17}	$(0,1,0,-1)$
v_{18}	$(0,1,0,1)$
v_{19}	$(1,0,-1,0)$
v_{20}	$(1,0,1,0)$
v_{21}	$(0,1,-1,0)$
v_{22}	$(0,1,1,0)$
v_{23}	$(1,0,0,-1)$
v_{24}	$(1,0,0,1)$

context	mutually orthogonal rays
C_1	$\{v_1, v_2, v_3, v_4\}$
C_2	$\{v_1, v_2, v_{15}, v_{16}\}$
C_3	$\{v_1, v_3, v_{19}, v_{20}\}$
C_4	$\{v_1, v_4, v_{21}, v_{22}\}$
C_5	$\{v_2, v_3, v_{23}, v_{24}\}$
C_6	$\{v_2, v_4, v_{17}, v_{18}\}$
C_7	$\{v_3, v_4, v_{13}, v_{14}\}$
C_8	$\{v_5, v_6, v_7, v_8\}$
C_9	$\{v_5, v_6, v_{14}, v_{16}\}$
C_{10}	$\{v_5, v_7, v_{18}, v_{20}\}$
C_{11}	$\{v_5, v_8, v_{21}, v_{23}\}$
C_{12}	$\{v_6, v_7, v_{22}, v_{24}\}$
C_{13}	$\{v_6, v_8, v_{17}, v_{19}\}$
C_{14}	$\{v_7, v_8, v_{13}, v_{15}\}$
C_{15}	$\{v_9, v_{10}, v_{11}, v_{12}\}$
C_{16}	$\{v_9, v_{10}, v_{13}, v_{16}\}$
C_{17}	$\{v_9, v_{11}, v_{17}, v_{20}\}$
C_{18}	$\{v_9, v_{12}, v_{21}, v_{24}\}$
C_{19}	$\{v_{10}, v_{11}, v_{22}, v_{23}\}$
C_{20}	$\{v_{10}, v_{12}, v_{18}, v_{19}\}$
C_{21}	$\{v_{11}, v_{12}, v_{14}, v_{15}\}$
C_{22}	$\{v_{13}, v_{14}, v_{15}, v_{16}\}$
C_{23}	$\{v_{17}, v_{18}, v_{19}, v_{20}\}$
C_{24}	$\{v_{21}, v_{22}, v_{23}, v_{24}\}$

7.9 Completed KCBS pentagon

Put $h = \sqrt{\cos(\pi/5)}$. The ten ray representatives used by the numerical branch are:

ray	representative
p_0	$(1, 0, h)$
p_1	$(\cos(4\pi/5), \sin(4\pi/5), h)$
p_2	$(\cos(8\pi/5), \sin(8\pi/5), h)$
p_3	$(\cos(12\pi/5), \sin(12\pi/5), h)$
p_4	$(\cos(16\pi/5), \sin(16\pi/5), h)$
$q_0 = p_0 \times p_1$	$(h[\sin(0) - \sin(4\pi/5)], h[\cos(4\pi/5) - \cos(0)], \sin(4\pi/5 - 0))$
$q_1 = p_1 \times p_2$	$(h[\sin(4\pi/5) - \sin(8\pi/5)], h[\cos(8\pi/5) - \cos(4\pi/5)], \sin(8\pi/5 - 4\pi/5))$

ray	representative
$q_2 = p_2 \times p_3$	$(h[\sin(8\pi/5) - \sin(12\pi/5)], h[\cos(12\pi/5) - \cos(8\pi/5)], \sin(12\pi/5 - 8\pi/5))$
$q_3 = p_3 \times p_4$	$(h[\sin(12\pi/5) - \sin(16\pi/5)], h[\cos(16\pi/5) - \cos(12\pi/5)], \sin(16\pi/5 - 12\pi/5))$
$q_4 = p_4 \times p_0$	$(h[\sin(16\pi/5) - \sin(0)], h[\cos(0) - \cos(16\pi/5)], \sin(0 - 16\pi/5))$

The five contexts are

$$\{p_0, q_0, p_1\}, \quad \{p_1, q_1, p_2\}, \quad \{p_2, q_2, p_3\}, \quad \{p_3, q_3, p_4\}, \quad \{p_4, q_4, p_0\}.$$

8 AI-assisted revision disclosure

OpenAI Codex with a GPT-5.6-based model was used in July 2026 to consolidate the audit program, compare its machine-readable outputs with the manuscript and supplement, and propose editorial revisions. The author specified the mathematical constructions, executed the reported run, inspected the program and outputs, and remains responsible for the sources, claims, and interpretation. Exact status is assigned only after rational primal and dual verification by the program.

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